

Effect of flapping mode on aerodynamic characteristics of bat-like flapping-wing aerial vehicle

Bosong Duan, Tianyou Mao, Kening Gong, Chuangqiang Guo* and Hong Liu

Abstract— In this paper, three reciprocating flapping modes of bats are analyzed and studied. According to the flapping structure of bat-like flapping-wing aerial vehicle (FAV), the structural modes corresponding to the three reciprocating flapping modes are established, that is, the symmetric flapping mode (S), the asymmetric flapping mode with the upward flapping angle greater than the downward flapping angle (UR) and the asymmetric flapping mode with the downward flapping angle greater than the upward flapping angle (U). The influence of the three flapping modes on the structural strength is verified by theoretical calculation. The fluid-structure interaction simulation (FSI) is used to verify the influence of three flapping modes on aerodynamic characteristics. Finally, the advantages and disadvantages of three flapping methods are compared and analyzed, it is concluded that under the same motion and flow field conditions, the pitching moment and thrust of UR are the smallest, and the lift is the largest, which is suitable for slow and stable flight; U has the largest thrust and pitching moment, and the smallest lift, which is conducive to rapid flight; The pitching moment, thrust and lift of S are moderate, and the performance is balanced, which is suitable for conventional flight. This study is helpful for the structural design of bat-like FAV.

I. INTRODUCTION

Since the 1990s, with the development of new materials and micro electro mechanical technology, Aerial vehicle has entered the trend of development, and all kinds of aerial vehicles have been studied all over the world [1]. The flight mode of the aerial vehicle includes fixed wing, rotor wing and flapping wing [2]. Compared with fixed wing and rotor wing, flapping wing is considered to be the optimal flight mode in biological evolution [3]. What's more, due to the bionic shape conditions, FAV has good invisibility, low noise and high maneuverability. It has great prospect in military investigation and scientific research [4].

Bat is known as the "flying master" of nature. It can not only fly fast, but also has better mobility than birds of the same

size, which is very consistent with the research requirements of MAV [5,6]. When bats fly, there are three reciprocating flapping modes according to different motion requirements. The angle between wing and body plane is defined as the flapping angle, the upward flapping angle is defined as the maximum flapping angle when the wing is above to the body, the downward flapping angle is defined as the maximum flapping angle when the wing is below to the body. So, the three flapping modes are S, UR and U. At present, most of the flapping modes of bat-like FAV adopt S flapping mode, such as the "Microbat" researched by University of California Institute of Technology [7], a memory alloy joint bat-like FAV researched by Brown University [8], and Bat Bot2 developed by the University of Illinois [9]. The large "Bionic Flying Fox" designed by German Festo has a wingspan of 228 cm and a weight of 580 g, this bat-like FAV adopts UR flapping mode. As for the U flapping mode, no public prototype has adopted it at present, but it can be found from the picture and video of bat flight that bats usually adopt U flapping mode when accelerating flight. Perhaps the different flapping modes of the bionic bats lead to different flapping modes of each bat-like FAV prototype, but different flapping modes must have different aerodynamic characteristics to meet different flight requirements.

In this paper, based on the flapping structure of bat-like FAV, the effects of these three flapping modes on the structure and aerodynamic characteristics are studied. Firstly, the bat-like FAV flapping mechanism was optimized to obtain the structural parameters that meet the corresponding flapping angle of three flapping modes, and the angular acceleration of three flapping modes was analyzed to obtain the influence of three flapping modes on the structural strength. Then, Abaqus and XFlow co-simulation were used to simulate the flow field of three flapping modes for the same wing, and the aerodynamic characteristics parameters under the same parameters were obtained. Finally, the aerodynamic characteristics of the three flapping modes were compared and analyzed, the advantages and disadvantages of different flapping modes are obtained, and the suitable application scenarios are summarized.

II. STRUCTURAL ANALYSIS OF THREE FLAPPING MODES

The existing flapping mechanism of FMAV mainly includes double crank-double rocker mechanism [11] and single crank-double rocker mechanism [12]. The mechanism of single crank-double rocker will cause a phase difference between the left and right wings in the process of synchronous flapping, which will lead to the instability of the FAV. Therefore, the double crank-double rocker mechanism is selected in this design to realize the synchronous reciprocating flapping of two wings. The schematic diagram of crank rocker mechanism with one wing is shown in Fig. 1.

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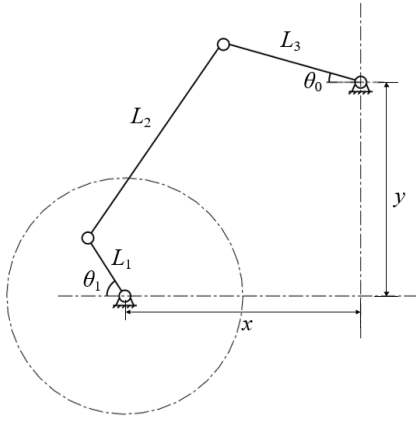


Fig.1 The crank rocker mechanism

In Fig. 1, L_1 is crank, L_2 is rocker, L_3 is flapping link, θ_1 is the crank angle, θ_0 is the flapping angle.

From the geometric relationship in the figure, it can be got:

$$\theta_1 \in (0, 2\pi), \theta_0 \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \quad (1)$$

$$(y - L_1 \sin \theta_1 + L_3 \sin \theta_0)^2 + (x + L_1 \cos \theta_1 - L_3 \cos \theta_0)^2 = L_2^2 \quad (2)$$

Assume that:

$$D_1 = x^2 + y^2 + L_1^2 - L_2^2 + L_3^2 + 2L_1(x \cos \theta_1 + y \sin \theta_1) \quad (3)$$

$$D_2 = -2L_3(y - L_1 \sin \theta_1) \quad (4)$$

$$D_3 = 2L_3(x + L_1 \cos \theta_1) \quad (5)$$

$$\varphi_1 = \arcsin \frac{D_2}{\sqrt{D_2^2 + D_3^2}} \quad (6)$$

The relationship between the flapping angle θ_0 and the crank angle θ_1 can be obtained:

$$\theta_0 = \varphi_1 \pm \arccos \frac{D_1}{\sqrt{D_2^2 + D_3^2}} \quad (7)$$

Through calculation, it is found that when the calculation formula takes the negative sign, the value range of θ_0 is $(0, \pi)$, which does not meet the conditions, while the calculation formula takes a positive sign, the value range of $(-\pi/2, \pi/2)$ meets the conditions, so the formula always takes a positive sign, that is:

$$\theta_0 = \varphi_1 + \arccos \frac{D_1}{\sqrt{D_2^2 + D_3^2}} \quad (8)$$

Equation (8) is the relation equation between flapping angle and parameters of each rod and crank angle. The crank angle θ_1 changes periodically at constant speed controlled by the motor, which is the fixed parameter $(0, 2\pi)$. So, the flapping angle is mainly affected by the rod parameters. The genetic-particle swarm hybrid optimization algorithm is used to optimize the rod parameters of the flapping mechanism with the minimum angular acceleration as the optimization objective. Three flapping modes correspond to three different constraints:

A. S flapping mode, flapping angle $(-27.5^\circ, 27.5^\circ)$

$$x = [x_1, x_2, x_3] = [L_1, L_2, L_3] \quad (9)$$

$$\text{Minf}(x) = \max(|\alpha(\theta_0)|) \quad (10)$$

s.t.

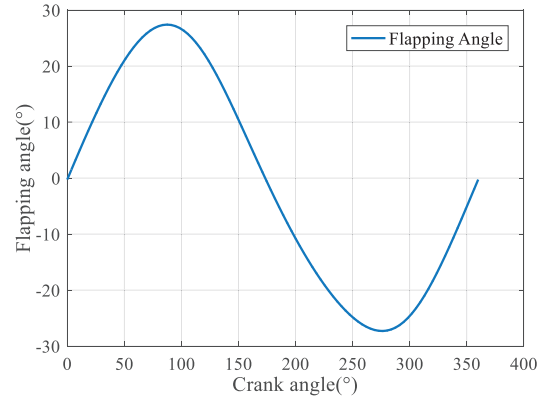
$$g_1 = |\max(\theta_0) - 27.5| \leq \text{delt_max} \quad (11)$$

$$g_2 = |\max(\theta_0) + \min(\theta_0)| \leq \text{delt_sym} \quad (12)$$

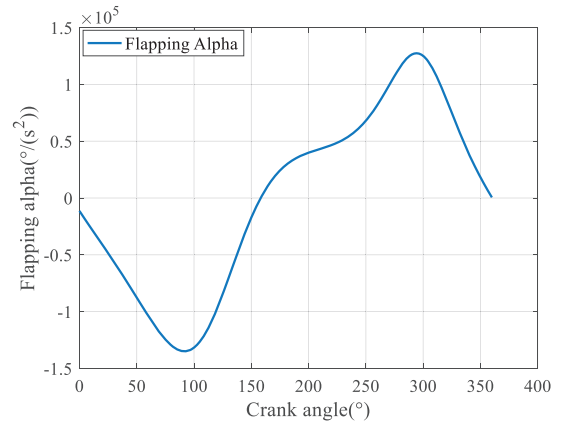
$$8 \leq x_1 \leq 10, 27 \leq x_2 \leq 35, 18 \leq x_3 \leq 22 \quad (13)$$

Where $\alpha(\theta_0)$ is the angular acceleration of flapping angle, delt_max represents the peak error accuracy, delt_sym represents the precision of symmetry error, g_1 ensures that the flapping angle is 27.5° , g_2 ensures the symmetry of the flapping angle.

Through optimization, the rod length parameter is $[L_1, L_2, L_3] = [8.0057, 31.1269, 18.0670]$. Corresponding flapping angle and angular acceleration curves are shown in Fig. 2. The value range of flapping angle $(-27.5^\circ, 27.5^\circ)$ meets the requirements of symmetric flapping and meets the constraint standards. From the angular acceleration curve, the maximum angular acceleration is $1.3488 \times 10^5 \text{ }^\circ/\text{s}^2$.



(a) Flapping angle



(b) Angular acceleration

Fig.2 The flapping angle and angular acceleration of S flapping mode

B. U flapping mode, flapping angle $(-20^\circ, 35^\circ)$

$$x = [x_1, x_2, x_3] = [L_1, L_2, L_3] \quad (14)$$

$$\text{Minf}(x) = \max(|\alpha(\theta_o)|) \quad (15)$$

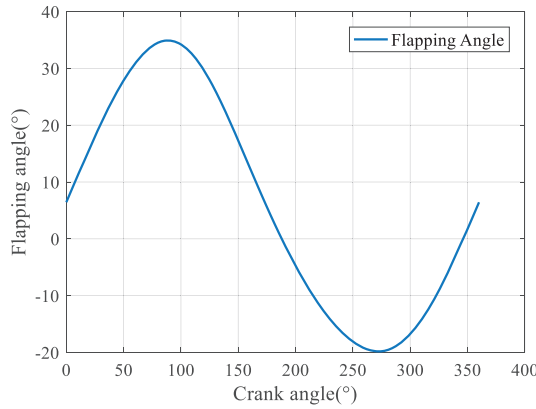
s.t.

$$g_1 = |\max(\theta_o) - 35| \leq \text{delt_max} \quad (16)$$

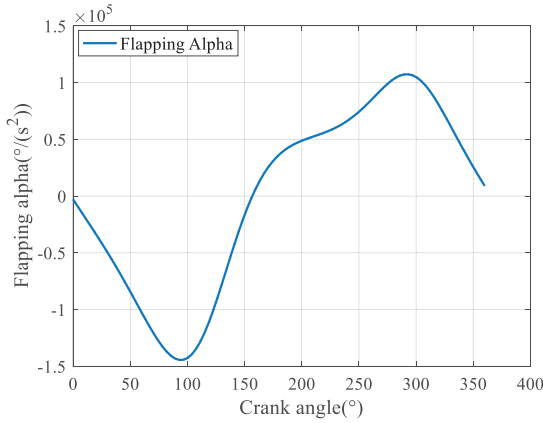
$$g_2 = |\max(\theta_o) + \min(\theta_o) - 15| \leq \text{delt_sym} \quad (17)$$

$$8 \leq x_1 \leq 10, 27 \leq x_2 \leq 35, 18 \leq x_3 \leq 22 \quad (18)$$

Through optimization, the rod length parameter is $[L_1, L_2, L_3] = [8.2211, 33.1335, 18.0205]$. Corresponding flapping angle and angular acceleration curves are shown in Fig. 3. The value range of flapping angle $(-20^\circ, 35^\circ)$ meets the requirements of unsymmetric flapping and meets the constraint standards. From the angular acceleration curve, the maximum angular acceleration is $1.4438 \times 10^5 \text{ }^\circ/\text{s}^2$.



(a) Flapping angle



(b) Angular acceleration

Fig.3 The flapping angle and angular acceleration of U flapping mode

C. UR flapping mode, flapping angle $(-35^\circ, 20^\circ)$

$$x = [x_1, x_2, x_3] = [L_1, L_2, L_3] \quad (19)$$

$$\text{Minf}(x) = \max(|\alpha(\theta_o)|) \quad (20)$$

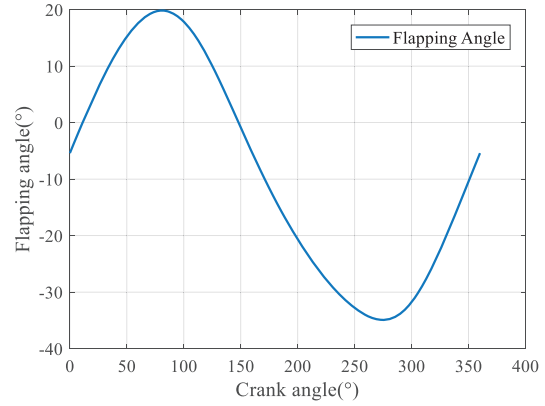
s.t.

$$g_1 = |\max(\theta_o) - 20| \leq \text{delt_max} \quad (21)$$

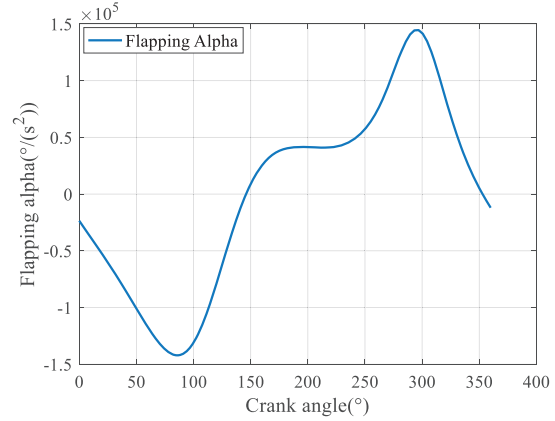
$$g_2 = |\max(\theta_o) + \min(\theta_o) + 15| \leq \text{delt_sym} \quad (22)$$

$$8 \leq x_1 \leq 10, 27 \leq x_2 \leq 35, 18 \leq x_3 \leq 22 \quad (23)$$

Through optimization, the rod length parameter is $[L_1, L_2, L_3] = [9.6004, 28.0028, 21.9976]$. Corresponding flapping angle and angular acceleration curves are shown in Fig. 4. The value range of flapping angle $(-35^\circ, 20^\circ)$ meets the requirements of unsymmetric flapping and meets the constraint standards. From the angular acceleration curve, the maximum angular acceleration is $1.4456 \times 10^5 \text{ }^\circ/\text{s}^2$.



(a) Flapping angle



(b) Angular acceleration

Fig.4 The flapping angle and angular acceleration of UR flapping mode

The smaller the maximum flapping angular acceleration is, the smaller the inertia moment of the transmission mechanism is, and the smaller the impact on the transmission structure is, which is beneficial to the stability of the transmission structure. From the angle of mechanism flapping, the maximum angular acceleration of symmetrical flapping is the smallest, which is most beneficial to the stability of transmission structure. In the actual flight process, different flapping modes not only affect the stability of flapping structure, but also affect the aerodynamic characteristics of wings, thus affecting the flight characteristics of FAUV. Therefore, the influence of different flapping modes on aerodynamic characteristics should be considered in this paper.

III. SIMULATION RESULTS AND ANALYSIS

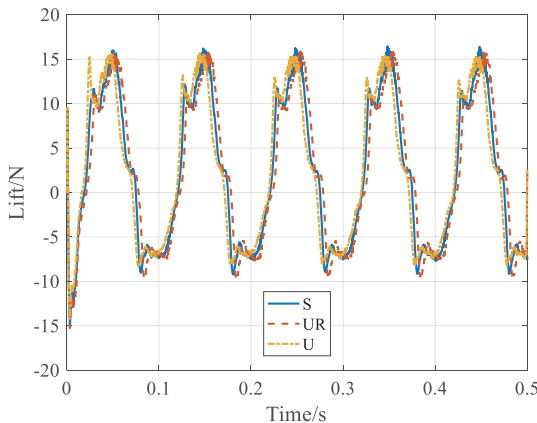
In order to verify the effects of three flapping modes on aerodynamic characteristics, Abaqus and XFlow co-simulation are used to calculate the aerodynamic force of

the wing. XFlow observes the change of flow field and aerodynamic force received by wing, and transmits the force received by wing to Abaqus. Abaqus observes the deformation of the wing and returns the deformation and movement of the wing to XFlow. The co-simulation of the two can better simulate the real situation of wing.

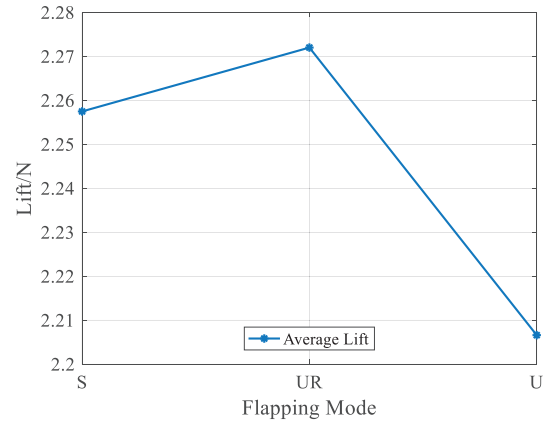
The simulation parameters are described in detail. The wing material is PDMS, this material density is 1000 kg/m^3 , the elastic modulus is 4 MPa , and the Poisson's ratio is 0.3 . Abaqus is used to mesh the model. The hexahedral mesh is used for the mesh, and C3D8R is selected for the mesh element, which is used to conduct three-dimensional stress analysis of the wing plane. In order to reduce the influence of boundary conditions on the flow field, the wing model is placed in a wind tunnel of about $10l \times 10l \times 15l$, and l is the length of bat wingspan. In the flow field area, Z-axis direction is set as freestream flow velocity, X-axis and Y-axis have no flow velocity, and the outlet boundary is also the freestream outflow with velocity gradient of 0 . To ensure the convergence of the results, set the Courant number to 1 . The particle size in the wind tunnel is set as $0.01l$, which is enough to ensure the accuracy of calculation. The flow field medium chooses the air under normal temperature, and the other simulation conditions are set as the flapping frequency is 10 Hz , the angle of attack of the wing is 15° , and the freestream flow velocity is 5 m/s .

The comparative curves of lift, thrust and pitching moment of different flapping modes are obtained by flow field simulation, which are shown in Fig. 5, Fig. 6 and Fig. 7.

Fig.5a shows the simulation lift of different flapping modes, Fig. 5b shows the average lift of different flapping modes. As the simulation does not reach a stable state in the first period, the data error is large, so the data in the last four periods are taken to calculate the average value. It can be seen from Fig. 5b that the average lift force of the three flapping modes is 2.2575 N , 2.2720 N and 2.2066 N . Statistically, the average lift of the three flapping modes is almost the same, but there are still some slight differences. By calculation, the lift generated by UR flapping mode is increased by 0.6% compared with S flapping mode, and by 3% compared with U flapping mode. From the perspective of lift alone, UR flapping mode is the best, S flapping mode is slightly inferior to UR and U is the worst.



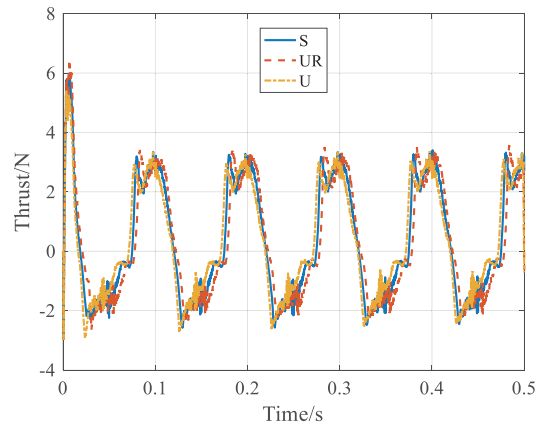
(a) Lift of three flapping modes



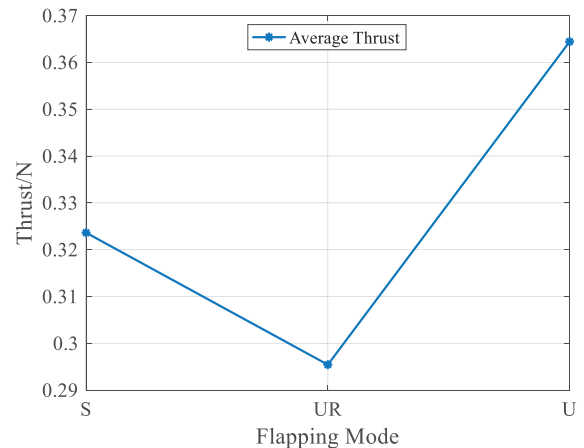
(b) Average Lift of three flapping modes

Fig.5 Lift and average lift corresponding to three flapping modes

Fig.6a shows the simulation thrust of different flapping modes, Fig. 6b shows the average thrust of different flapping modes. It can be seen from Fig. 6b that the average thrust of the three flapping modes is 0.3236 N , 0.2954 N and 0.3644 N . By calculation, the thrust generated by U flapping mode is increased by 12.6% compared with S flapping mode, and by 23.3% compared with UR flapping mode. From the perspective of thrust alone, U flapping mode is the best, S flapping mode is slightly inferior to U and UR is the worst.



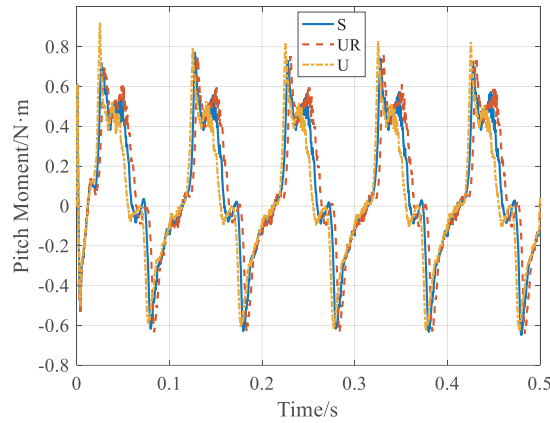
(a) Thrust of three flapping modes



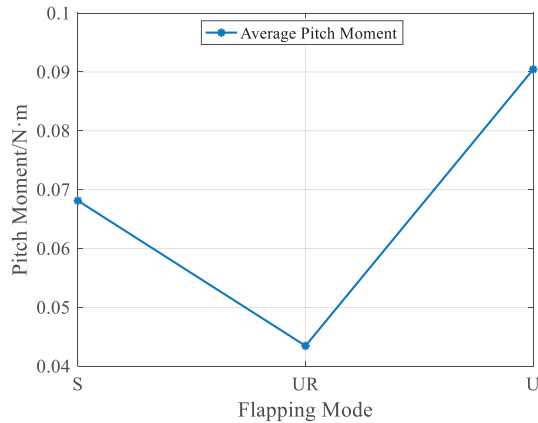
(b) Average thrust of three flapping modes

Fig.6 Thrust and average thrust corresponding to three flapping modes

Fig.7a shows the simulation pitch moment of different flapping modes, Fig. 7b shows the average pitch moment of different flapping modes. It can be seen from Fig. 7b that the average pitch moment of the three flapping modes is 0.0681 N·m, 0.0435 N·m and 0.0904 N·m. By calculation, the pitch moment generated by UR flapping mode is decreased by 36% compared with S flapping mode, and by 52% compared with U flapping mode. From the perspective of pitch moment alone, UR flapping mode is the best, S flapping mode is slightly inferior to UR and U is the worst.



(a) Pitch moment of three flapping modes



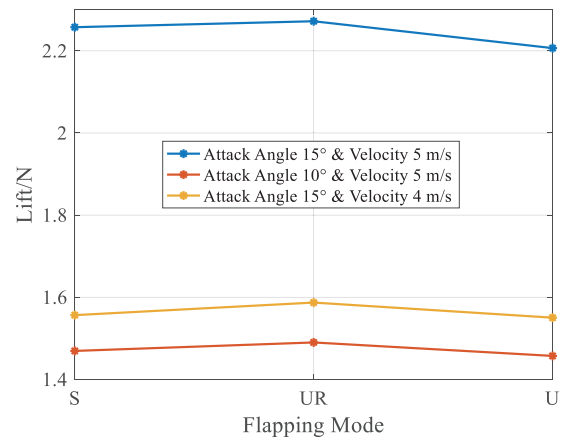
(b) Average pitch moment of three flapping modes

Fig.7 Pitch moment and average pitch moment corresponding to three flapping modes

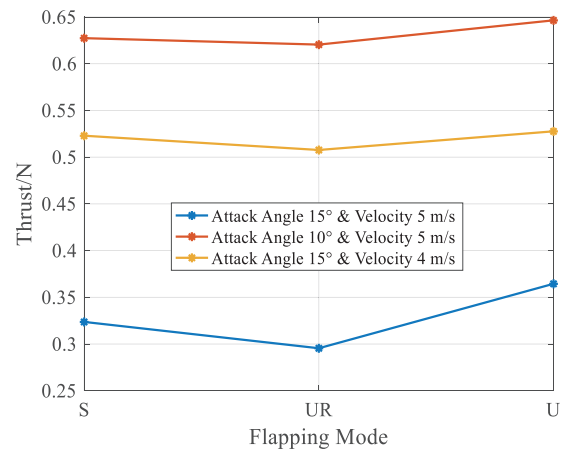
In summary, the three flapping modes have almost the same effect on the lift. The UR flapping mode reduces the pitching moment, but reduces the thrust. U generates great thrust, but the pitching moment is also too large. S flapping mode is between UR flapping mode and U flapping mode, which can guarantee certain thrust and control pitching moment at the same time.

The conclusion of a single working condition may be accidental, which is not sufficient to explain the accuracy of the conclusion. The angle of attack and freestream flow velocity will be changed to calculate the impact of the three

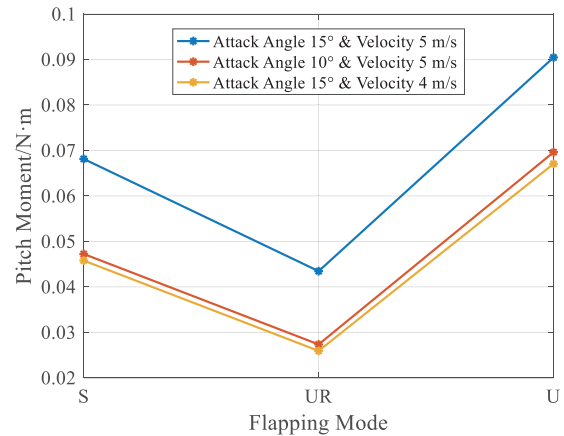
flapping modes on aerodynamic characteristics under different working conditions, as shown in Fig. 8.



(a) Average lift moment of three flapping modes



(b) Average thrust of three flapping modes



(c) Average pitch moment of three flapping modes

Fig.8 Average aerodynamic characteristic of three flapping modes under different working conditions

Fig.8 shows the average aerodynamic characteristic of three flapping modes under different working conditions. It can be seen from Fig. 8 that the specific values of the curve are different, which is due to the different aerodynamic forces

under different working conditions. However, the variation law of the curve is the same, which indicates that under different working conditions, the influence of the three kinds of flapping modes on the aerodynamic force is the same, so the conclusions obtained above are universal. Through the calculation of average aerodynamic force under different working conditions, it can be found that the lift generated by UR flapping mode is increased by 1.3 % compared with S flapping mode, and 2.5 % compared with U flapping mode. The thrust produced by U flapping mode is increased by 5.5% and 10.5% compared with S flapping mode and UR flapping mode. Compared with S flapping mode, the pitching moment produced by UR flapping mode is reduced by 40.6 % and increased by 58 %.

In summary, UR flapping mode can significantly reduce the pitching moment, which is beneficial to improving the stability of flapping. However, this flapping mode will sacrifice a lot of thrust, which is not conducive to fast flight. Therefore, UR flapping mode is suitable for FAV with large weight and high stability requirements. U flapping mode can greatly improve the thrust, but the pitching moment is too large, resulting in poor flight stability, so U flapping mode is suitable for small FAV with certain requirements for speed. S flapping mode is between UR flapping mode and U flapping mode, which can ensure certain thrust and control the pitching moment. The performance of S flapping mode is the most balanced, so S flapping mode is suitable for conventional FAV.

IV. CONCLUSION

In this paper, the effects of three flapping modes on the aerodynamic characteristics of the bat-like FAV are studied, and the advantages and disadvantages of the three flapping modes are analyzed. Based on the structure of double crank rocker driving mechanism, the maximum angular acceleration of three flapping modes is calculated. Through comparative analysis, it is found that the maximum angular acceleration of S flapping mode is the smallest, and its structural stability is the best. The maximum angular acceleration of UR and U flapping modes are almost the same, and the structural stability is slightly worse. At the same time, Abaqus and XFlow are used to simulate the flow field of the three flapping modes. By comparing the aerodynamic forces acting on the wings, it is found that the UR flapping mode can significantly reduce the pitching moment and is conducive to the stability of flight, but the UR flapping mode will sacrifice a lot of thrust and is not conducive to high-speed flight. U flapping mode can greatly improve thrust, but excessive pitching moment will lead to flight instability; S flapping mode is between UR flapping mode and U flapping mode, which can ensure a certain thrust and control the pitching moment at the same time. It is a relatively balanced and neutral flapping mode. Combined with the advantages and disadvantages of the three flapping modes, this study believes that the UR flapping mode is suitable for large FAV with large weight and high stability requirements, the U flapping mode is suitable for small FAV with certain speed requirements, and the S flapping mode is suitable for conventional FAV. This conclusion is conducive to the design of flapping structure of FAV.

In the future, a FAV will be designed, which can automatically adjust the flapping mode and improve flight performance according to different flight requirements.

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